

Color Atlas of Dental Medicine

Editors:
Klaus H. Rateitschak
Herbert F. Wolf

Radiology

Friedrich A. Pasler



Thieme

Table of Contents

Radiographic Examination of the Patient in the Dental Office

- 3 Examination Strategy and Active Protection Against Radiation Exposure**
- 4 Examination Strategies**

Panoramic Radiography for Basic Information and Supplemental Examination Using Special Radiographs

- | | |
|--|---|
| 9 Technique for Panoramic Radiography | 47 The Bite-Wing Radiograph |
| 10 Technique for Panoramic Radiography | 50 Examples of Diagnosis Using Bite-Wings |
| 12 Positioning of the Patient in the Apparatus | |
| 14 Increased Radiographic Quality through Positioning According to Indication | 51 Apical and Periodontal Radiographic Technique |
| 17 Typical Incorrect Positioning | 52 Radiographic Surveys for Patients of Various Ages |
| 18 Incorrect Positioning | 53 Apical and Periodontal Survey in Adults |
| 19 Positioning in the Mixed Dentition Stage | 54 Maxilla |
| 20 Positioning to Visualize Periodontal Destruction | 58 Mandible |
| 21 Positioning of the Tongue | 62 Third Molars |
| 22 Depiction of the Alveolar Ridges | |
| 23 "Zonarc," A Special Instrument for Clinics | 63 Radiographic Technique with the Occlusal Film |
| 24 Special Radiographs Using the Cephalometric Attachment | 64 Intraoral |
| | 68 Extraoral |
| 25 Radiographic Anatomy in the Panoramic Radiograph | 69 Radiographic Anatomy in Periapical and Occlusal Radiographs |
| 26 Survey of the Anatomic Structures Visible in a Panoramic Radiograph | 70 Radiographic Anatomy During Tooth Development |
| 27 Ventral Portion of the Facial Skeleton | 71 Radiographic Anatomy of Special Regions |
| 28 Ventral Portion of the Facial Skeleton in the Maxilla | 71 Maxillary Anterior Region |
| 29 Variations in the Maxillary Sinus | 72 Maxillary Canine Region |
| 30 Retromaxillary Space | 73 Maxillary Premolar Region |
| 31 External Ear and Temporomandibular Joint Region | 75 Maxillary Molar Region |
| 32 Palatal Bone in the Shadow of the Coronoid Process | 76 Mandibular Anterior Region |
| 33 Tuberosity Region and the Cervical Vertebrae | 77 Mandibular Canine Region |
| 34 Chin Region | 78 Mandibular Premolar Region |
| 36 Chin Region and the Body of the Mandible | 80 Mandibular Molar Region |
| 37 Mandibular Canal, Mandibular Rami and the Cervical Vertebrae | 82 Radiographic Anatomy in Occlusal Radiographs |
| 38 Mandibular Canal and Retromolar Structures | |
| 39 Hyoid Bone and Cervical Region | 83 Localization Using Various Methods |
| 40 Hyoid Bone and Subtraction Effect from the Base of the Skull | 86 Panoramic Radiography as an Aid in Localization |
| 41 Angle of the Mandible and the Styloid Process | 87 Special Localization Problems |
| 42 Examination of Children and Adolescents Using the Panoramic Radiograph | 87 Buccally Impacted Mandibular Canine |
| 45 Diagram of Formation and Eruption of the Deciduous Teeth | 88 Ectopically Positioned Anterior Teeth |
| 46 Diagram of Formation and Eruption of the Permanent Teeth | 89 Apically Displaced Maxillary Third Molars |
| | 90 Axially Presented Maxillary Third Molars |

91	Errors in Technique That Reduce Radiograph Quality	110	Conventional Skull Films
92	Tips for Preparation of Good Panoramic Radiographs	110	First Standard Projection: Posteroanterior Cephalometric Skull Projection
92	Before Positioning the Patient in the Apparatus	111	Second Standard Projection: Lateral Cephalometric Skull Projection
93	With the Patient in the Apparatus	112	Special Construction of a High Capacity Cephalometric Instrument
94	Tips for Preparation of Good Radiographs	113	Third Standard Projection: Full Axial Projection of the Skull
94	Before Exposing the Film	114	Lateral Oblique Projection of the Mandible
95	Immediately Before Exposure	115	Special Positioning for the Lateral Oblique Projection of Mandible
96	Summary of Basic Rules for Preparation of High Quality Radiographs	116	Zygoma/Cheek Tangential Skull Projections
97	Common Errors During Preparation of Periapical Radiographs	117	Mandibular Posteroanterior Skull Radiograph (Reverse Towne)
102	Common Errors During Preparation of Occlusal Radiographs	118	Waters' Projection Radiograph
103	Processing Technique and Errors Leading to Poor Quality Radiographs	119	Supplemental Examination of the Maxillary Sinus with Additional Methods
104	Tips for Error-Free Processing	120	Transcranial Projection: "The Modified Schüller TMJ Film, Open and Closed"
105	Tips for Error-Free Development	121	Tomography: Supplemental Examination of the Temporomandibular Joint (TMJ)
106	Tips for Error-Free Fixation	122	Computed Tomography (CT)
107	Optimum Radiograph	124	Magnetic Resonance Imaging
108	Reducing Overdeveloped Radiographs		
109	Supplemental Examinations Using Conventional and Modern Imaging Techniques		

Selected Examples of Dental Radiographic Diagnosis

127	Anomalies of Dental Development and the Teeth	154	Sclerosing Osteomyelitis and Inflammatory Reactive Osteosis
128	Congenitally Missing Teeth, Retention and Inclusion	155	Osteomyelitis in Infants and Children
129	Retention, Malocclusion and Resorption	156	Acute Osteomyelitis
130	Retention of Supernumerary Teeth, Resorption of Retained Teeth	157	Secondary Chronic Osteomyelitis
131	Retained Teeth in Special Locations	158	Primary Chronic Osteomyelitis, Osteoradionecrosis
132	Retained and Ankylosed Teeth	159	Dentogenic Sinus Disorders
133	Mesiodens, Gemination, Taurodontism and Dens in dente	160	Panoramic Radiography of Dentogenic Sinus Pathology
134	Hypercementosis and Enamel Pearls	162	Additional Signs of Dentogenic Infection
135	Amelogenesis Imperfecta	163	Incidental Findings and the Significance of the Waters' Projection
136	Dentinogenesis Imperfecta	164	Acute Unilateral Dentogenic Sinusitis
137	Additional Dental Dysplasias	165	Schematic View of Sinus Diagnosis in Waters' Projection Radiographs
138	Posterior Open Bite with Macroglossia and "Idiopathic Root Resorption"	167	Acute and Chronic Maxillary Sinusitis
139	Concrements, Calcifications, Ossifications	168	Computed Tomography for Supplemental Diagnosis
147	Regressive Changes in Teeth and Jaws	170	Foreign Bodies, Root Fragments and Surgical Defects
151	Inflammation of the Jaws and Osteoradionecrosis	171	Temporomandibular Joint Disturbances
152	Acute and Chronic Apical Periodontitis	172	Examination of the Masticatory Organ Using the Panoramic Radiograph
153	Diffuse Sclerosing Osteomyelitis in Chronic Apical and Marginal Periodontitis	173	Temporomandibular Joint Pain with Malocclusion